Nome _____ Cognome _____ Data ____/____/_____

Thermal physics: mock test

Time: 40 minutes. Total: 30 marks. Calculator, ruler and sharp pencil allowed. Equations are not given; data and constants are at the end of Part B. In Parts A and C circle one letter for each question. In Part B write only the final answer on the line (there is some space for working if you want it).

A Multiple choice [12]

Choose one letter for each question. Each question is worth 1 mark.

1. A thermometer reads 2 °C in melting ice and 98 °C in boiling water, both at standard atmospheric pressure. Its scale is evenly divided. What is the true temperature when it reads 26 °C? **[1]**

A 24 °C
B 25 °C
C 26 °C
D 28 °C

2. A flat metal ring has a circular hole in the centre. The ring is heated evenly. What happens to the diameter of the hole? **[1]**

A It increases by the same fraction as every other length of the ring.
B It stays the same, because the hole is empty space.
C It increases, but by less than the metal expands.
D It decreases, because the metal expands inwards.

3. A metal rod is heated and its temperature rises by 20 °C. Its length increases by 2.0 mm. By how much would its length increase if its temperature rose by 60 °C? **[1]**

A 2.0 mm
B 4.0 mm
C 6.0 mm
D 18 mm

4. A solid is heated with a constant power. While it is melting, its temperature stays constant. What happens to the energy supplied? **[1]**

A It increases the average kinetic energy of the particles.
B It is lost to the surroundings.
C It makes the particles larger.
D It is used to overcome the forces between the particles, so their potential energy increases.

5. Which statement is true of evaporation, but not of boiling? **[1]**

A It happens only at the surface, at any temperature below the boiling point.
B It happens only at one fixed temperature.
C Bubbles of vapour form throughout the liquid.
D It happens only when the liquid is heated.

6. A fixed mass of gas in a sealed rigid container is cooled. The pressure of the gas falls. Why? **[1]**

A The particles lose mass.
B The particles move further apart.
C The particles move more slowly.
D The particles become smaller.

7. Two metal blocks, P and Q, have the same mass. Each block is given the same amount of energy. The temperature of P rises by 10 °C and the temperature of Q rises by 25 °C. Which statement is correct? **[1]**

A P has a smaller specific heat capacity than Q.
B P has a larger specific heat capacity than Q.
C P and Q have the same specific heat capacity.
D P received more energy than Q.

8. A metal spoon and a wooden spoon are left in hot soup. The handle of the metal spoon soon becomes hot but the wooden handle does not. What is the main reason? **[1]**

A Metals have a larger specific heat capacity than wood.

B Metals expand more when they are heated.

C The metal radiates more infrared radiation to the hand.

D Free electrons carry energy quickly through the metal.

9. A room is cooled by an air-conditioning unit fixed near the ceiling. Why is this position effective? **[1]**

A Cold air is denser than warm air, so it sinks and warm air rises to take its place, setting up a convection current.

B Cold air is less dense than warm air, so it rises and spreads through the room.

C Air is a good conductor, so energy passes quickly through the whole room.

D Radiation from the unit is strongest downwards.

10. Energy from the Sun reaches the Earth through space. Which process transfers it, and why? **[1]**

A Conduction, because particles in space pass on the energy.

B Radiation, because it does not need a medium.

C Radiation, because the energy is carried by particles of air.

D Convection, because hot gas rises from the Sun to the Earth.

11. 200 g of hot water is mixed with 300 g of cold water in an insulated cup. No energy is lost to the surroundings. Which statement is correct? **[1]**

A The final temperature is exactly halfway between the two starting temperatures.

B The temperature fall of the hot water equals the temperature rise of the cold water.

C The energy lost by the hot water equals the energy gained by the cold water.

D The cold water gains more energy than the hot water loses, because there is more of it.

12. A hot object is left to cool in a cold room. Which statement about its cooling is correct? **[1]**

A It cools at a constant rate until it reaches room temperature.

B It cools below room temperature before levelling out.

C It cools fastest at the end, when it is almost at room temperature.

D It cools fastest at the start, when the difference between its temperature and the room temperature is largest.

B Short answers **[12]**

Write the final answer on the line, with its unit. Each question is worth 2 marks.

13. A steel bridge section is 80 m long at 10 °C. Calculate the increase in its length when its temperature rises to 35 °C. Give your answer in millimetres. **[2]**

Answer _____ mm

14. An electric kettle has a power of 2.5 kW. Calculate the time needed to heat 1.2 kg of water from 20 °C to 100 °C. Assume that no energy is lost. **[2]**

Answer _____ s

15. A heater of power 100 W warms a 0.40 kg metal block from 20 °C to 45 °C in 90 s. All the energy from the heater goes into the block. Calculate the specific heat capacity of the metal. **[2]**

Answer _____ J/(kg °C)

16. A heater of power 500 W melts 0.40 kg of ice at 0 °C. All the energy from the heater is absorbed by the ice. Calculate the time taken to melt all the ice. **[2]**

Answer _____ s

17. 0.30 kg of water at 90 °C is mixed with 0.20 kg of water at 15 °C in an insulated container. Calculate the final temperature. **[2]**

Answer _____ °C

18. A steel rod is 2.0 m long. When it is heated its length increases by 0.60 mm. Calculate the rise in temperature of the rod. **[2]**

Answer _____ °C

Data and constants

Specific heat capacity of water	4200 J/(kg °C)
Specific latent heat of fusion of ice	3.3×10^5 J/kg
Linear expansion coefficient of steel	12×10^{-6}/°C
Temperature in kelvin	$T/\mathrm{K} = \theta/^\circ\mathrm{C} + 273$
Law of cooling	$T - T_{\mathrm{room}} = (T_0 - T_{\mathrm{room}})\, e^{-kt}$, so $\ln(T - T_{\mathrm{room}}) = \ln(T_0 - T_{\mathrm{room}}) - kt$

C Alternative to practical [6]

First experiment. A group calibrates an Arduino temperature sensor against a liquid-in-glass reference thermometer. The scale of the reference thermometer is divided in 1 °C divisions. The sensor gives a number N (in counts).

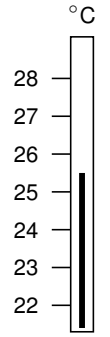


Fig. 1

19. The reference thermometer reads 1.0 °C in melting ice. It is then placed in a water bath and the liquid level is as shown in Fig. 1. What is the correct temperature of the bath, with its reading uncertainty? **[1]**

A (25.5 ± 0.5) °C
B (24.5 ± 0.5) °C
C (24.5 ± 1.0) °C
D (26.5 ± 0.5) °C

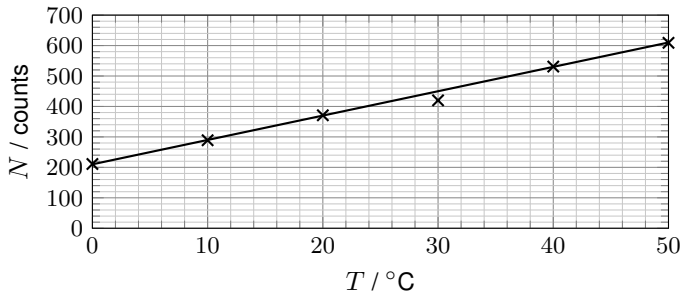


Fig. 2

20. Fig. 2 shows N against T for six baths, with the best-fit line. At which temperature is the point anomalous? **[1]**

A 0 °C
B 20 °C
C 30 °C
D 40 °C

21. What is the gradient of the best-fit line in Fig. 2? **[1]**

A 0.125 counts/°C
B 8.0 counts/°C
C 12.2 counts/°C
D 210 counts/°C

Second experiment. A potato is taken out of a hot water bath at a time that only the teacher knows. A group presses Start ($t = 0$) at 10:05 and records the temperature of the potato for 20 min. The room temperature is $T_{\text{room}} = 21.0$ °C and the bath was at 45.0 °C. The law of cooling is on the data page.

t / s	T / °C	$(T - T_{\text{room}})$ / °C	$\ln(T - T_{\text{room}})$
0	41.1	20.1	3.001
300	38.3	17.3	2.851
600	35.9	14.9	2.701
900	33.8	12.8	2.549
1200	32.0	11.0	2.398

Table 1

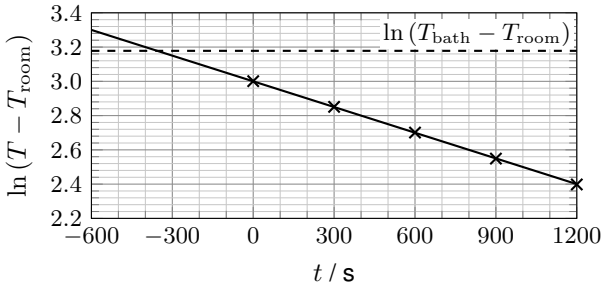


Fig. 3

22. Which graph should the group plot to obtain a straight line? **[1]**

A $\ln(T - T_{\text{room}})$ against t
B T against t
C T against $\ln t$
D $(T - T_{\text{room}})$ against t

23. Use the best-fit line in Fig. 3. What is the value of k in the law of cooling? **[1]**

A 2.0×10^3 s
B 0.60/s
C -5.0×10^{-4}/s
D 5.0×10^{-4}/s

24. The dashed line is at $\ln(T_{\text{bath}} - T_{\text{room}})$. At what time was the potato taken out of the bath? **[1]**

A 09:45
B 09:53
C 09:59
D 10:11